

Unifying Special Relativity and Quantum Mechanics

Lectures 20–21 — Interactions Between Matter and Gravity, and Gravity as an Effective Field Theory

(The equivalence principle, the vierbein, the linear coupling $-\frac{\kappa}{2}h_{\mu\nu}T^{\mu\nu}$, and why quantum gravity breaks down at the Planck scale)

Typed version of the handwritten notes, with added explanations in blue boxes

How to read this document

Conventions. Exactly as in Lectures 1–19: normal text and the yellow / pink / grey boxes are a faithful transcription of the handwritten sheets, and the text colours follow the pens of the original (blue, magenta, green, red, pencil). Everything inside a blue box like this one is material I have added; it is *not* in the original notes.

Pagination. This scan opens a *new* chapter, “Interactions between matter and gravity”, and the page numbering **restarts at 1**. Since the numbering of the previous chapter carried nine stars, these sheets are marked with ten: [p. 1*****]–[p. 26*****].

Two things about this scan.

- (a) *Two sheets out of order.* In the scanned file, page p. 4***** comes *before* page p. 3*****. The logical order is $2 \rightarrow 3 \rightarrow 4 \rightarrow 5$, and that is the order used here: p. 3***** introduces the scalar action and the first two steps of the recipe, p. 4***** completes step 2 in freely-falling coordinates. The sentence that runs across the break — “...the equivalence principle tells us that, in order to / immerse the scalar field in a gravitational field ...” — confirms the ordering. If you follow the paper scan, expect to flip forward one sheet at that point and back again.
- (b) *Two lectures, one page series.* There is no page break or title between them, but the material divides cleanly at [p. 12*****], where the red heading “What goes wrong, and why, with quantum gravity?” opens a new subject. I have therefore split this document into LECTURE 20 (pp. 1–11) and LECTURE 21 (pp. 12–26). The split is my inference from the content, not something marked in the notes.

Slips flagged in place. Three small ones: the boxed definition of the inverse tetrad on [p. 7*****] has a clashing index (b where a is meant); the Planck length quoted on [p. 26*****] is 1.68×10^{-35} m where the standard value is 1.62×10^{-35} m; and the general dimension formula on [p. 13*****] writes both particle counts with the same letter. Each is corrected and flagged where it occurs.

Where we left off — recap of Lecture 19

Lecture 19 completed the field-theoretic construction of gravity. Exchanging one graviton between two static masses gave Newton’s potential, $E_0 = -\kappa^2 M_1 M_2 / 32\pi r$ with $\kappa^2 = 32\pi G_N$, attractive because the mediator has even spin and masses are positive. It then showed that linearized gravity is *inconsistent*: coupling $h_{\mu\nu}$ to a matter energy–momentum tensor destroys the conservation of that very tensor, and the attempt to repair this order by order forces an infinite tower of self-interactions together with a deformed gauge symmetry. That deformed

symmetry turned out to be general coordinate invariance, with

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + \kappa h_{\mu\nu}(x),$$

and the resummed action the Einstein–Hilbert action.

These two lectures pick up both loose ends. LECTURE 20 asks: given that gravity *is* the metric, how does ordinary matter couple to it? The answer comes from the equivalence principle and a mechanical recipe, and at linear order it reproduces exactly the coupling $-\frac{\kappa}{2}h_{\mu\nu}T^{\mu\nu}$ that Lectures 17–19 had been assuming. LECTURE 21 then asks what happens when this theory is treated as a quantum field theory — and finds that it fails, in a specific and instructive way, at the Planck scale.

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LECTURE 20 — Interactions between matter and gravity (pp. 1*****–11*****)

Interactions between matter and gravity

1 Introduction: the equivalence principle [p. 1*****]

[p. 1*****]

■ Introduction

Consider Newton's equation for a particle in a constant gravitational field

$$m_I \frac{d^2 \vec{r}}{dt^2} = m_G \vec{g}$$

↙ inertial mass

↗ gravitational mass

■ $\vec{r} = \vec{r}(t)$ position vector of the particle■ $\vec{g}$ is the acceleration due to the external constant gravitational field

↓ If inertial and gravitational masses are the same $m_I = m_G \equiv m$

$$m \frac{d^2}{dt^2} \left[\vec{r}(t) - \frac{1}{2} \vec{g} t^2 \right] = 0$$

Consequently, if we set $\vec{r}'(t) \equiv \vec{r}(t) - \frac{1}{2} \vec{g} t^2$ we get

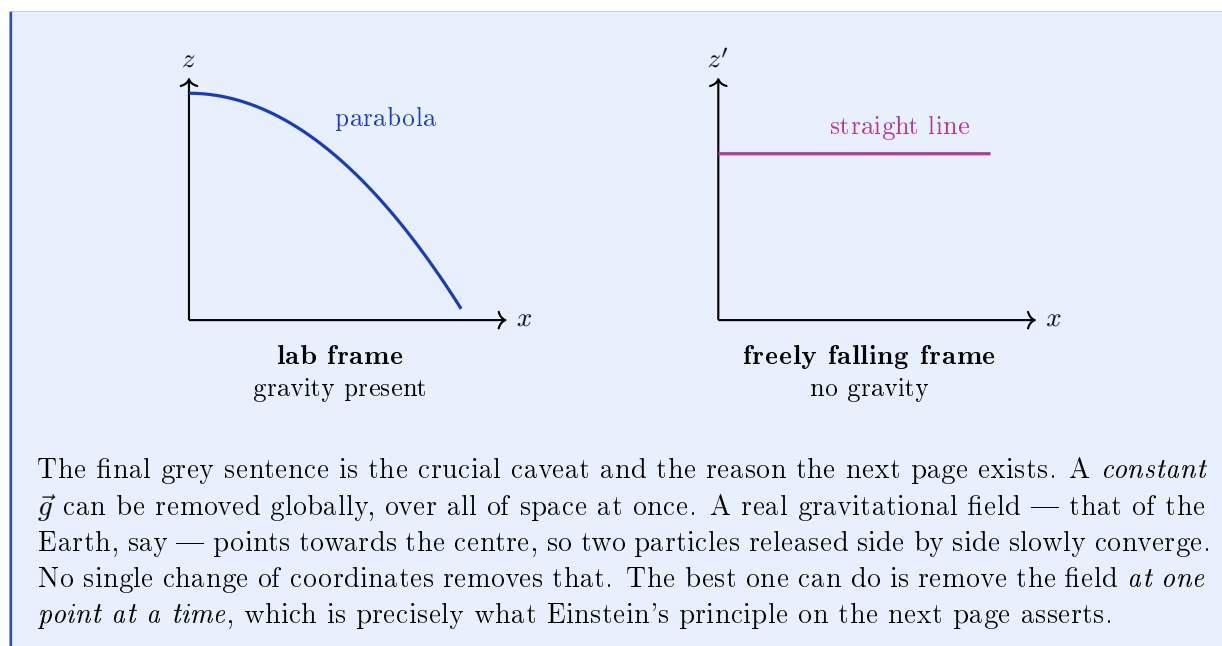
$\frac{d^2 \vec{r}'(t)}{dt^2} = 0 \quad \rightsquigarrow \quad$ when viewed from a freely falling frame the particle experience no gravity

Of course, we were able to do this because the external gravitational field was constant, but in general, gravitational fields are not constant.

What this one line is really saying

The equation $m_I \ddot{\vec{r}} = m_G \vec{g}$ contains two logically independent quantities. m_I measures how hard it is to accelerate the body; m_G measures how strongly gravity pulls on it. Nothing in Newtonian mechanics requires them to be equal — for the electric force, the analogous “charge” Q is manifestly unrelated to m_I , which is exactly why different charges fall differently in an electric field.

Experimentally $m_I = m_G$ to about one part in 10^{15} (Eötvös, and modern torsion-balance and satellite tests). Once they are set equal, the mass cancels entirely from the equation of motion: *every* body falls the same way. And if every body falls the same way, one can simply travel along with them — the substitution $\vec{r}' = \vec{r} - \frac{1}{2} \vec{g} t^2$ is nothing but “jump”.



2 Einstein’s principle, and the recipe [p. 2*****]

[p. 2*****]

To account for this, Einstein formulated the following principle:

Gravitational fields are of such a nature that, at each point in space-time, they allow themselves to be transformed away by choosing an appropriate set of coordinates

↔► Of course, this set of coordinates will vary from space-time point to space-time point, but there will always be a set of coordinates in terms of which it looks like there’s no gravitation.

“Recipe” for including gravitation:

- 1) Take any local quantity, such as a Lagrangian density or an infinitesimal volume element, written in a Lorentz-invariant way so as to satisfy the laws of special relativity
- 2) Identify the coordinates appearing in the local quantities with the “freely-falling” coordinates
- 3) When expressed in terms of arbitrary space-time variables, the interaction with gravity will be magically generated

Why the recipe works, and what “magically” is doing

The three steps are less mysterious than they look. The principle guarantees that at any chosen point X there exist coordinates in which physics is exactly special-relativistic. So:

- **Step 1** says: start from a law you already trust in flat space. The restriction to *local* quantities is essential — the principle only removes gravity at a point, so only statements made at a point can be carried over.
- **Step 2** says: those flat coordinates are not arbitrary labels; they are the freely-falling ones at that point.
- **Step 3** says: now describe the same physics in whatever coordinates you actually use. The

transformation between the two carries the information about the gravitational field, and re-expressing the flat-space law in the new variables automatically produces the coupling to gravity.

Nothing is really magic: the “magic” is that the Jacobian matrix relating the two coordinate systems — which is about to be named the *vierbein* — is exactly the object out of which the metric is built. The recipe never postulates a coupling; it derives one. And the pay-off, on [p. 11*****], is the coupling $-\frac{\kappa}{2}h_{\mu\nu}T^{\mu\nu}$ that was simply *assumed* in Lectures 17–19. This procedure is standard and goes by the name *minimal coupling*: in practice it amounts to $\eta_{\mu\nu} \rightarrow g_{\mu\nu}$ and $d^4x \rightarrow \sqrt{-g}d^4x$, which is exactly the result reached on [p. 9*****].

3 The real scalar field: local quantities [p. 3*****]

[p. 3*****]

We use latin indices $a, b, c, \dots$ in the freely falling frame, and greek indices in arbitrary coordinates

■ The real scalar field

Consider a real scalar field $\phi(x)$ described, in the absence of gravity, by the action

$$S = \int d^4x \left[\frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - V(\phi) \right]$$

$$\partial_\mu = \frac{\partial}{\partial x^\mu} = (\partial_t, \vec{\nabla})$$

$$\partial^\mu = \eta^{\mu\nu} \partial_\nu = (\partial_t, -\vec{\nabla}) ; \quad \eta^{\mu\nu} \text{ the flat metric}$$

$$\blacktriangleright V(\phi) = \frac{m^2}{2}\phi^2 + \frac{\lambda}{4!}\phi^4$$

We apply the procedure defined before.

1) Local quantities

The local quantities are the Lagrangian density

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - V(\phi)$$

and the volume element $d^4x = dx^0 d\vec{x}$

2) Introduction of “freely-falling” coordinates

The equivalence principle tells us that, in order to immerse the scalar field in a gravitational field, [p. 4*****] we have to interpret the variable x^μ and its derivatives as being the “free-fall” coordinates

We thus identify

$$\{x^\mu\} \longrightarrow \{\xi^a\}_{a=0,1,2,3}$$

with ξ^a flat coordinates such that

$$ds^2 = \eta_{ab} d\xi^a d\xi^b$$

The Lagrangian is then

$$\mathcal{L} = \frac{1}{2} \eta^{ab} (\partial_a \phi) (\partial_b \phi) - V(\phi) \quad \text{with } \partial_a \equiv \frac{\partial}{\partial \xi^a}$$

In a similar way, the volume element d^4x becomes in terms of the flat coordinates

$$d^4x \longrightarrow d^4\xi = d\xi^0 d\xi^1 d\xi^2 d\xi^3$$

The action is

$$S = \int d^4\xi \left[\frac{1}{2} \eta^{ab} (\partial_a \phi) (\partial_b \phi) - V(\phi) \right]$$

Lagrangian in freely falling coord.

The index convention, and why it matters

Two alphabets are now in use, and keeping them straight is the whole game:

index		meaning
ξ^a	latin a, b, c	locally inertial (freely-falling) frame at a point; raised/lowered with η_{ab}
x^μ	greek μ, ν, ρ	arbitrary coordinates on the manifold; raised/lowered with $g_{\mu\nu}$

The action written on this page looks exactly like the flat-space one — and that is the point. Step 2 of the recipe is a *relabelling*, not a modification. Physically it says: an observer in free fall at a point sees the scalar field obeying ordinary special-relativistic physics. Gravity has not disappeared; it has been hidden. The next page says exactly where.

4 Non-inertial coordinates and the vierbein [p. 5*****]

[p. 5*****]

Obs. The presence of gravity in the above Lagrangian is secretly contained in the fact that the coordinates $\{\xi^a\}$ vary from point to point

↔ To be more precise, let us write $\{\xi_{\underline{X}}^a\}$

3) Introduction of a non-inertial coordinate system

To make the presence of gravity manifest, we introduce a non-inertial coordinate system $\{x^\mu\}$ over the space-time manifold

↔ We then introduce the vierbein (or tetrad)

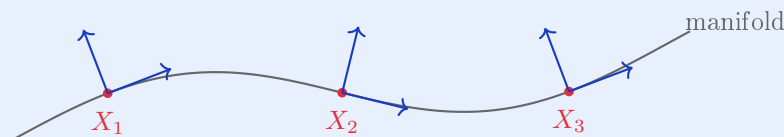
$$e^a{}_\mu(\underline{X}) \equiv \left. \frac{\partial \xi_{\underline{X}}^a(x)}{\partial x^\mu} \right|_{x=\underline{X}}$$

“pointwise Jacobian” the transformation matrix that, at each point, relates arbitrary space-time coordinates to locally inertial coordinates

The picture behind the notation ξ_X^a

The subscript X is the entire content of the observation in magenta. There is not one freely-falling coordinate system; there is a *different one at every point*, each valid only in an infinitesimal neighbourhood of its own point.

each point carries its own $\{\xi_X^a\}$



The vierbein e^a_μ is the dictionary between the two descriptions at a point. “Vierbein” is German for “four legs” (*tetrad* in Greek): the four vectors e^a_μ , $a = 0, 1, 2, 3$, are the four legs of the local inertial frame planted at that point. In matrix language it is just the Jacobian $\partial\xi^a/\partial x^\mu$ — but a Jacobian of a transformation that is only defined at one point, which is why it cannot be integrated into a global change of coordinates. *That failure to integrate is the gravitational field.*

5 The metric from the tetrad [p. 6*****]

[p. 6*****]

With this definition the metric in the arbitrary coordinate system x^μ , evaluated at point $\underline{X}$, is

$$g_{\mu\nu}(\underline{X}) = \left. \frac{\partial \xi_X^a(x)}{\partial x^\mu} \right|_{x=\underline{X}} \left. \frac{\partial \xi_X^b(x)}{\partial x^\nu} \right|_{x=\underline{X}} \eta_{ab} = e^a_\mu(\underline{X}) e^b_\nu(\underline{X}) \eta_{ab}$$

Since $\underline{X}$ was arbitrary, the same construction can be performed at every point of the coordinate patch.

We thus promote the pointwise tetrad $e^a_\mu(\underline{X})$ to a tetrad field, $e^a_\mu(x)$

↔ With this definition, we write

$$g_{\mu\nu}(x) = e^a_\mu(x) e^b_\nu(x) \eta_{ab}$$

$$\star \quad e^a_\mu = \frac{\partial \xi^a}{\partial x^\mu}$$

△ With the understanding that ξ^a denotes a locally inertial frame at point x and not a single global coordinate system

In this formalism, space-time indices are raised/lowered with $g_{\mu\nu}$ while “locally inertial” indices with η_{ab}

We thus have:

The metric was there all along

This is the first genuinely new equation of the lecture. Read it as a definition of the metric *in terms of* the freely-falling frames:

$$g_{\mu\nu}(x) = e^a_\mu(x) e^b_\nu(x) \eta_{ab}.$$

It says the metric is the flat Minkowski metric, seen through the distorting lens of the point-dependent Jacobian. Equivalently: $g_{\mu\nu}$ tells you how to measure lengths in your coordinates, given that lengths are measured by η_{ab} in the local inertial frame.

Counting. The tetrad $e^a{}_\mu$ has $4 \times 4 = 16$ components; the metric it produces has only 10 (symmetric 4×4). The mismatch of 6 is exactly the dimension of the Lorentz group: a local Lorentz rotation $e^a{}_\mu \rightarrow \Lambda^a{}_b(x) e^b{}_\mu$ leaves $g_{\mu\nu}$ unchanged, because $\Lambda^a{}_c \Lambda^b{}_d \eta_{ab} = \eta_{cd}$. This is the freedom to choose how the local inertial observer at each point is oriented and boosted — of course the metric cannot depend on that.

Why bother with tetrads at all, when the metric alone would do? Two reasons. Here, it is the natural output of the equivalence-principle recipe: it *is* the free-fall Jacobian. More generally, tetrads are indispensable for coupling *spinors* to gravity — the Dirac equation needs γ matrices carrying flat indices, so it needs a local frame, not just a metric. That is beyond this lecture, but it is why the formalism is worth setting up.

6 Inverse tetrad and index manipulation [p. 7*****]

[p. 7*****]

■ Starting from $e^a{}_\mu(x)$, we define

$$e_b{}^\nu(x) = \eta_{ab} e^a{}_\mu(x) g^{\mu\nu}(x)$$

A clashing index in this box

As written, the index b appears on the left-hand side *and* twice on the right (once summed against η_{ab}), while a is left free on the right only. The intended definition, which is what the very next line actually uses, is

$$e_a{}^\nu(x) = \eta_{ab} e^b{}_\mu(x) g^{\mu\nu}(x)$$

i.e. “lower the latin index with η , raise the greek index with g ”. With this reading everything below is consistent.

↪ From which we see that

$$\begin{aligned} e^a{}_\mu(x) e_a{}^\nu(x) &= e^a{}_\mu(x) \eta_{ab} e^b{}_\rho(x) g^{\rho\nu}(x) = \\ &= \left(e^a{}_\mu(x) e^b{}_\rho(x) \eta_{ab} \right) g^{\rho\nu}(x) = g_{\mu\rho}(x) g^{\rho\nu}(x) = \delta_\mu^\nu \end{aligned}$$

► $e_a{}^\mu(x)$ is the inverse tetrad, and we write

$$e^a{}_\mu(x) e_a{}^\nu(x) = \delta_\mu^\nu$$

$$e_a{}^\mu(x) = \frac{\partial x^\mu}{\partial \xi^a}$$

■ Raising indices, we write

$$g^{\mu\nu}(x) = e_a{}^\mu(x) e_b{}^\nu(x) \eta^{ab}$$

■ For the derivative operator, we write

$$\partial_a = \frac{\partial}{\partial \xi^a} = \frac{\partial x^\mu}{\partial \xi^a} \partial_\mu = e_a^\mu \partial_\mu \quad \rightsquigarrow \quad \boxed{\partial_a = e_a^\mu \partial_\mu}$$

Everything on this page is one idea

The tetrad is an invertible 4×4 matrix, and e_a^μ is its inverse:

$$e^a_\mu e_a^\nu = \delta_\mu^\nu, \quad e^a_\mu e_b^\mu = \delta_b^a.$$

Once that is established, the tetrad becomes a machine for converting greek indices into latin ones and back, and every formula on the page is an application:

flat-frame object	same thing in arbitrary coordinates
η_{ab}	$g_{\mu\nu} = e^a_\mu e^b_\nu \eta_{ab}$
η^{ab}	$g^{\mu\nu} = e_a^\mu e_b^\nu \eta^{ab}$
∂_a	$\partial_a = e_a^\mu \partial_\mu$

The last line is just the chain rule, and it is the one that will be used immediately: it is how the derivatives in the freely-falling Lagrangian get converted into derivatives in the coordinates we actually use.

7 The volume element: where $\sqrt{-g}$ comes from [p. 8*****]

[p. 8*****]

■ Consider the volume element $d^4\xi$. By definition of the tetrad,

$$d^4\xi = |\det(e^a_\mu)| d^4x$$

From $g_{\mu\nu} = e^a_\mu e^b_\nu \eta_{ab}$ we write $g_{\mu\nu} = (e^T)_\mu^a \eta_{ab} e_b^\nu$, that is, in matrix form $g = e^T \eta e$

We thus write

$$\begin{aligned} \det(g_{\mu\nu}) &= \det(e^T \eta e) = \det(e)^2 \det \eta = (-1)(\det e)^2 \\ &\rightsquigarrow |\det(e)| = \sqrt{-\det(g_{\mu\nu})} = \sqrt{-g} \end{aligned}$$

We thus have

$$d^4\xi = \sqrt{-g} d^4x$$

As far as the Lagrangian is concerned, we write

$$\begin{aligned} \mathcal{L} &= \frac{1}{2} \eta^{ab} (\partial_a \phi) (\partial_b \phi) - V(\phi) = \frac{1}{2} \eta^{ab} e_a^\mu (\partial_\mu \phi) e_b^\nu (\partial_\nu \phi) - V(\phi) = \\ &= \frac{1}{2} \underbrace{(\eta^{ab} e_a^\mu e_b^\nu)}_{g^{\mu\nu}} (\partial_\mu \phi) (\partial_\nu \phi) - V(\phi) \end{aligned}$$

Two lines, two famous results

The measure. $\det \eta = -1$ in four dimensions with signature $(+, -, -, -)$: the product of the diagonal entries is $(+1)(-1)(-1)(-1) = -1$. That single minus sign is the entire reason the invariant measure is $\sqrt{-g}$ rather than $\sqrt{g}$ — the determinant of a Lorentzian metric is negative, so one takes the square root of $-g$ to get a real number.

Geometrically, $d^4\xi$ is the volume measured by the local inertial observer, which is the physically meaningful one; d^4x is a coordinate volume, which depends on how you happened to label points. $\sqrt{-g}$ is the conversion factor. (Compare polar coordinates in flat space: $d^2x = r dr d\theta$, where $r = \sqrt{\det \bar{g}}$ for $g = \text{diag}(1, r^2)$.)

The Lagrangian. The two inverse tetrads collide and become $g^{\mu\nu}$, so the flat-frame kinetic term turns into the curved-space one with no work at all. Notice what happened to $V(\phi)$: nothing. A potential has no derivatives and no indices, so minimal coupling does not touch it.

Together these two lines are the whole of “minimal coupling”:

$$\eta^{\mu\nu} \rightarrow g^{\mu\nu}, \quad d^4x \rightarrow \sqrt{-g} d^4x.$$

8 The action of a scalar field in a gravitational field, and the expansion of $g^{\mu\nu}$ [p. 9*****]

[p. 9*****]

We find the action of a scalar field in a gravitational field

$$S = \int d^4x \sqrt{-g} \left(\frac{1}{2} g^{\mu\nu} (\partial_\mu \phi) (\partial_\nu \phi) - V(\phi) \right)$$

Interactions at the linear order

To extract the interactions, we expand the metric by writing

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + \kappa h_{\mu\nu}(x)$$

Let us expand $g^{\mu\nu}(x)$ and $\sqrt{-g}$

■ Expansion of $g^{\mu\nu}(x)$

By definition $g^{\mu\nu} g_{\nu\rho} = \delta_\rho^\mu$

We make the ansatz $g^{\mu\nu} = \eta^{\mu\nu} + \kappa A^{\mu\nu}$, and write

$$\begin{aligned} g^{\mu\nu} g_{\nu\rho} &= (\eta^{\mu\nu} + \kappa A^{\mu\nu}) (\eta_{\nu\rho} + \kappa h_{\nu\rho}) = \delta_\rho^\mu + \kappa \eta^{\mu\nu} h_{\nu\rho} + \kappa \eta_{\nu\rho} A^{\mu\nu} + \mathcal{O}(\kappa^2) \\ &= \delta_\rho^\mu + \kappa (h^\mu{}_\rho + A^\mu{}_\rho) + \mathcal{O}(\kappa^2) \stackrel{!}{=} \delta_\rho^\mu \quad \rightsquigarrow \quad A^\mu{}_\rho = -h^\mu{}_\rho \end{aligned}$$

$$g^{\mu\nu} = \eta^{\mu\nu} - \kappa h^{\mu\nu} + \mathcal{O}(\kappa^2)$$

A useful sanity check

The result is the tensor version of the elementary expansion $(1+x)^{-1} = 1 - x + x^2 - \dots$: the inverse of $\eta + \kappa h$ is $\eta - \kappa h$ to first order. Two points worth keeping straight:

- $g^{\mu\nu}$ is defined as the *matrix inverse* of $g_{\mu\nu}$, not as $g_{\mu\nu}$ with indices raised — if it were the latter, the statement would be circular. The sign flip is the content.
- Indices on $h^{\mu\nu}$ are raised with η , not g . At the order we work this makes no difference, since the correction would be $\mathcal{O}(\kappa^2)$, but at higher orders the distinction matters.

9 Expansion of $\sqrt{-g}$ [p. 10***]**

[p. 10*****]

■ Expansion of $\sqrt{-g}$

$$\begin{aligned}
 g \equiv \det(g_{\mu\nu}) &= \det(\eta_{\mu\nu} + \kappa h_{\mu\nu}) = \\
 &= \det(\eta_{\mu\rho} \delta_\nu^\rho + \kappa \eta_{\mu\rho} h^\rho{}_\nu) = \det\left[\eta_{\mu\rho}(\delta_\nu^\rho + \kappa h^\rho{}_\nu)\right] = \\
 &= \det(\eta_{\mu\rho}) \det(\delta_\nu^\rho + \kappa h^\rho{}_\nu) = -\det(\delta_\nu^\rho + \kappa h^\rho{}_\nu) \\
 &\rightsquigarrow \boxed{-g = \det(\delta_\nu^\rho + \kappa h^\rho{}_\nu)}
 \end{aligned}$$

We use $\det M = \exp(\text{tr} \log M) \rightsquigarrow \det(\mathbf{1} + A) = \exp(\text{tr} \log(\mathbf{1} + A))$

Since $\log(\mathbf{1} + A) = A - \frac{1}{2}A^2 + \frac{1}{3}A^3 + \dots$

$$\text{tr} \log(\mathbf{1} + A) = \text{tr}\left(A - \frac{1}{2}A^2 + \frac{1}{3}A^3 + \dots\right) = \text{tr} A - \frac{1}{2}\text{tr}(A^2) + \dots$$

and

$$\exp(\text{tr} \log(\mathbf{1} + A)) = 1 + \text{tr} A + \mathcal{O}(A^2) \rightsquigarrow \det(\mathbf{1} + A) = 1 + \text{tr} A$$

And, in our case

$$-g = 1 + \kappa h + \mathcal{O}(\kappa^2) \rightsquigarrow \sqrt{-g} = (1 + \kappa h + \mathcal{O}(\kappa^2))^{1/2}$$

$$\sqrt{-g} = 1 + \frac{\kappa}{2}h + \mathcal{O}(\kappa^2)$$

The log det trick

The identity $\det M = \exp(\text{tr} \log M)$ is worth having permanently: it converts a determinant (hard to expand) into a trace (easy to expand). It is obvious in a basis where M is diagonal — the determinant is the product of eigenvalues, the trace of the log is the sum of their logs, and exp of a sum of logs is a product — and since both sides are basis independent, it holds generally.

Here $A = \kappa h^\rho{}_\nu$ and $\text{tr} A = \kappa h^\rho{}_\rho \equiv \kappa h$, the trace of the graviton field. So at first order the determinant only knows about the *trace* of $h_{\mu\nu}$; the traceless part enters at $\mathcal{O}(\kappa^2)$. Then $\sqrt{1+x} \simeq 1 + x/2$ gives the $\kappa h/2$.

That factor of $\frac{1}{2}$ is not decoration — it is what produces the $-\eta^{\mu\nu}\mathcal{L}$ piece of the energy-momentum tensor on the next page. Without it the coupling would not assemble into $T^{\mu\nu}$ at all.

10 The linear interaction: $-\frac{\kappa}{2}h_{\mu\nu}T^{\mu\nu}$ [p. 11*****]

[p. 11*****]

We now write for the Lagrangian

$$\sqrt{-g} \left(\frac{1}{2} g^{\mu\nu} (\partial_\mu \phi)(\partial_\nu \phi) - V(\phi) \right)$$

↓ at order $\mathcal{O}(\kappa)$

$$\left(1 + \frac{\kappa}{2} h \right) \left[\frac{1}{2} (\eta^{\mu\nu} - \kappa h^{\mu\nu}) (\partial_\mu \phi)(\partial_\nu \phi) - V(\phi) \right]$$

↓ we extract the interactions

$$\begin{aligned} \frac{\kappa}{2} h_{\mu\nu} \eta^{\mu\nu} \times \left[\frac{1}{2} (\partial_\alpha \phi)(\partial^\alpha \phi) - V(\phi) \right] - \frac{\kappa}{2} h_{\mu\nu} (\partial^\mu \phi)(\partial^\nu \phi) &= \\ = -\frac{\kappa}{2} h_{\mu\nu} \underbrace{\left\{ (\partial^\mu \phi)(\partial^\nu \phi) - \eta^{\mu\nu} \left[\frac{1}{2} (\partial_\rho \phi)(\partial^\rho \phi) - V(\phi) \right] \right\}}_{\text{canonical energy-momentum in flat space}} &= -\frac{\kappa}{2} h_{\mu\nu} T^{\mu\nu} \end{aligned}$$

We thus reconstruct the linear interaction

$$\mathcal{L}_{int}^{(h)} = -\frac{\kappa}{2} h_{\mu\nu} T^{\mu\nu}(\phi)$$

This is the punchline of Lecture 20

Where each piece came from. Two sources contribute at $\mathcal{O}(\kappa)$:

origin	contribution
$\sqrt{-g} \rightarrow 1 + \frac{\kappa}{2} h$ multiplying the whole $\mathcal{L}$	$+\frac{\kappa}{2} h_{\mu\nu} \eta^{\mu\nu} \left[\frac{1}{2} (\partial\phi)^2 - V \right]$
$g^{\mu\nu} \rightarrow \eta^{\mu\nu} - \kappa h^{\mu\nu}$ in the kinetic term	$-\frac{\kappa}{2} h_{\mu\nu} (\partial^\mu \phi)(\partial^\nu \phi)$

Factor out $-\frac{\kappa}{2} h_{\mu\nu}$ and what is left in the braces is precisely the canonical energy-momentum tensor of the free scalar, $T^{\mu\nu} = (\partial^\mu \phi)(\partial^\nu \phi) - \eta^{\mu\nu} \mathcal{L}$. Neither piece alone would give $T^{\mu\nu}$; it is the combination — and hence the factor $\frac{1}{2}$ in $\sqrt{-g}$ — that makes it work.

Why this matters. Lectures 17–19 *assumed* that gravity couples to the energy-momentum tensor with universal strength κ , and built everything on it. Lecture 20 has now *derived* that coupling from the equivalence principle alone, with no reference to spin-2 fields, propagators, or soft theorems. The two routes — “massless spin-2 field, made consistent” and “special relativity, made locally valid at every point” — meet exactly here.

A remark on generality. Nothing in the derivation used the fact that ϕ is a scalar with this particular potential. Any matter Lagrangian, run through the same recipe, produces $-\frac{\kappa}{2} h_{\mu\nu} T^{\mu\nu}$ with its own $T^{\mu\nu}$. That is universality, obtained for free.

LECTURE 21 — Gravity as an effective field theory (pp. 12*****–26*****)

11 Graviton-mediated scattering [p. 12*****]

[p. 12*****]

■ What goes wrong, and why, with quantum gravity?

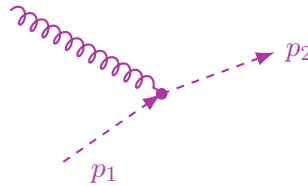
Consider the linearized theory

$$\mathcal{L} = \mathcal{L}_0 + \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{\kappa}{2}h_{\mu\nu}T^{\mu\nu}(\phi)$$

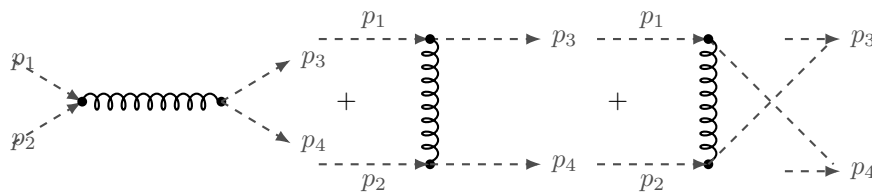
$$\kappa^2 = 32\pi G_N = \frac{32\pi}{M_P^2}$$

and compute the amplitude for the scattering $S(p_1) + S(p_2) \rightarrow S(p_3) + S(p_4)$ mediated by gravity. We work at the tree level.

The linear coupling $-\frac{\kappa}{2}h_{\mu\nu}T^{\mu\nu}(\phi)$ gives an interaction vertex of the kind



And the scattering proceeds via the diagrams



Reading the three diagrams

The dashed lines are the scalar S , the coiled line the graviton. Each vertex carries a factor κ , so every diagram is $\mathcal{O}(\kappa^2)$ — that is the counting used two pages later.

The three diagrams are the s -, t - and u -channels: the two incoming particles can annihilate into a virtual graviton (s), or particle 1 can emit a graviton absorbed by particle 2 with p_1 flowing into p_3 (t), or the same with p_3 and p_4 exchanged (u). The t -channel is the one that, in the static non-relativistic limit, produced Newton's potential in Lecture 19; here we are asking about the opposite regime, $E \rightarrow \infty$.

“Tree level” means no loops, so no renormalization subtleties enter. The problem found below is not a divergence of loop integrals — it is present already in the classical, tree-level

amplitude.

12 The mass dimension of a scattering amplitude [p. 13*****—p. 15*****]

[p. 13*****]

Let's compute the amplitude without doing computations.

We need the following result.

The scattering amplitude $i\mathcal{M}$ for a $2 \rightarrow 2$ scattering process is dimensionless in natural units
More in general, we have

$$[i\mathcal{M}(n \rightarrow m)] = [M]^{4-n-m}$$

A notation point

The original writes both particle counts with the same letter, which makes the exponent read $4 - m - m$. From the derivation that follows — where the initial state has n particles and the final state m — the intended statement is

$$[i\mathcal{M}(n \rightarrow m)] = [M]^{4-n-m},$$

and this is confirmed by the boxed result $\Delta_{\mathcal{M}} = 4 - n - m$ on [p. 15*****]. For $2 \rightarrow 2$ this gives $[M]^0$: dimensionless, as stated.

Let's prove this result

We start from the scattering amplitude



$$\underbrace{\langle \vec{K}_1, \dots, \vec{K}_m | S | \vec{P}_1, \dots, \vec{P}_n \rangle}_{\text{we elaborate on this term}} = (2\pi)^4 \delta(P_1 + \dots + P_n - K_1 - \dots - K_m) i\mathcal{M}$$

We elaborate on this term. We remind that

$$\begin{cases} |\vec{K}\rangle = (2E_k)^{1/2} a^\dagger(\vec{K})|0\rangle \\ [a(\vec{K}), a^\dagger(\vec{P})] = (2\pi)^3 \delta(\vec{K} - \vec{P}) \end{cases} \quad \begin{array}{l} \text{Remember that } [\delta(\vec{P})] = [M]^{-3} \text{ since} \\ \int d^3\vec{P} \delta(\vec{P}) = 1 \end{array}$$

We thus extract

$$[a(\vec{P})] = [a^\dagger(\vec{P})] = [M]^{-3/2}$$

And, as a consequence, we have

$$[|\vec{K}\rangle] = [M]^{-1}$$

[p. 14*****]

For a multi-particle states, we have

$$[|\vec{P}_1, \dots, \vec{P}_n\rangle] = [M]^{-n}$$

Furthermore, we also have

$$S = \sum_{m=0}^{\infty} \frac{(-i)^m}{m!} \int d^4x_1 \cdots d^4x_m T[\mathcal{H}_{int}(x_1) \cdots \mathcal{H}_{int}(x_m)]$$

$\Leftrightarrow$ that implies

$$[S] = [M]^0$$

So that

$$[\langle \vec{K}_1, \dots, \vec{K}_m | S | \vec{P}_1, \dots, \vec{P}_n \rangle] = [M]^{-n-m}$$

We now move to the RHS of ★

$$\left[(2\pi)^4 \delta(K_1 + \cdots + K_m - P_1 - \cdots - P_n) i\mathcal{M} \right] = [M]^{\Delta_{\mathcal{M}} - 4}$$

↑ mass-dimension of $i\mathcal{M}$ ↑ from the δ -function

[p. 15*****]

From the comparison, we find

$$\Delta_{\mathcal{M}} = 4 - n - m$$

The bookkeeping, made explicit

Every step is a dimension count; here they are collected.

object	mass dimension
$\delta^{(3)}(\vec{P})$	-3 (since $\int d^3P \delta = 1$ and $[d^3P] = +3$)
$a(\vec{P}), a^\dagger(\vec{P})$	-3/2 (from $[a, a^\dagger] = (2\pi)^3 \delta^{(3)}$)
$ \vec{K}\rangle = (2E)^{1/2} a^\dagger 0\rangle$	-3/2 + 1/2 = -1
$ \vec{P}_1 \dots \vec{P}_n\rangle$	-n
S	0 (it is exp of a dimensionless exponent)
$\delta^{(4)}(\sum p)$	-4

Setting $[\text{LHS}] = [\text{RHS}]$: $-n - m = \Delta_{\mathcal{M}} - 4$, hence $\Delta_{\mathcal{M}} = 4 - n - m$.

Why $[S] = 0$. $S = T \exp(-i \int d^4x \mathcal{H}_{int})$. A Hamiltonian density has dimension 4 and $[d^4x] = -4$, so the exponent is dimensionless — as it must be, since one cannot exponentiate a dimensionful quantity. Every term in the sum then has dimension zero.

The result to remember: for $2 \rightarrow 2$, $\Delta_{\mathcal{M}} = 0$. A $2 \rightarrow 2$ amplitude is a pure number. This innocuous fact is about to do all the work.

13 $i\mathcal{M} \sim E^2/M_P^2$, and the loss of unitarity [p. 15*****–p. 16*****]

We come back to our computation.

If we only count the powers of couplings in the diagrams we would have

$$i\mathcal{M} = \mathcal{O}(\kappa^2)$$

► since $[\kappa^2] = [M]^{-2}$, but $i\mathcal{M}$ must be dimensionless, we need something to compensate

The only dimensionfull quantity we can use is the energy (all particles are massless)

We thus find

$$i\mathcal{M} = \mathcal{O}(E^2 \kappa^2) = \mathcal{O}\left(\frac{E^2}{M_P^2}\right)$$

△ The scattering amplitude grows with energy

[p. 16*****]

The fact that a scattering amplitude grows with energy signals a problem

↔► Intuitively, a scattering amplitude is a probability amplitude. It cannot grow indefinitely with energy because it eventually gives probabilities bigger than 1.

The theory is no longer unitary when the scattering amplitude for a 2-to-2 process becomes

$$i\mathcal{M} \gtrsim \mathcal{O}(1)$$

↔ The quantum theory of gravitational interactions “breaks down” and violate unitarity at energies

$$E \gtrsim \mathcal{O}(M_P)$$

What is the physical significance of this effect?

↔ We note that this bad behaviour is crucially connected to the fact that the interaction is non-renormalizable

[p. 17*****]

The argument in three moves, and what it does and does not say

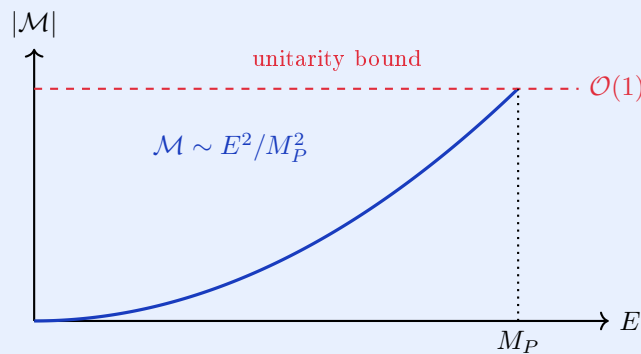
1. Counting. Each graviton vertex carries κ , and the tree diagrams have two vertices: $i\mathcal{M} \propto \kappa^2$.

2. Restoring dimensions. $[\kappa^2] = [M]^{-2}$, but the amplitude must be dimensionless. Something with dimension $[M]^{+2}$ must appear. At energies far above all masses, the centre-of-mass energy is the only scale in the problem, so it must be E^2 :

$$i\mathcal{M} \sim \kappa^2 E^2 = \frac{32\pi E^2}{M_P^2}.$$

No Feynman integrals were computed — dimensional analysis alone fixes the energy dependence.

3. Unitarity. $S^\dagger S = \mathbf{1}$ bounds partial-wave amplitudes: probabilities cannot exceed one. An amplitude growing without bound must therefore be wrong, and the estimate says it goes wrong at $E \sim M_P \sim 10^{19}$ GeV.



What this does *not* mean. It does not mean general relativity is wrong, or that its predictions at ordinary energies are suspect — at $E \ll M_P$ the amplitude is fantastically small and the theory works beautifully. It means the theory is *incomplete*: something must replace or supplement it before M_P is reached. Distinguishing “wrong” from “incomplete” is exactly what the rest of this lecture is about.

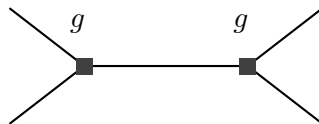
14 Non-renormalizable interactions: the general logic [p. 17*****]

Consider the following logic.

Suppose that we have a non-renormalizable interaction



► If we compute a 2-to-2 scattering amplitude



At high-energy, where we neglect all masses $E \gg m$, the energy is the only scale that can compensate the g^2 powers of coupling

$$i\mathcal{M} = \mathcal{O}(g^2 E^2)$$

(assuming $[g] = [M]^{-1}$)

$\hookrightarrow$ Breaking of unitarity at some energy scale

In order to gain some understanding, let us explore a specific, simple, toy model.

Renormalizable versus non-renormalizable, by dimensions

The classification is nothing more than the mass dimension of the coupling. Since $[\mathcal{L}] = 4$, an operator $\mathcal{O}$ of dimension d enters as $g\mathcal{O}$ with $[g] = 4 - d$:

$[g]$	operator dim.	name	behaviour at high E
> 0	$d < 4$	relevant (super-renormalizable)	dies away
$= 0$	$d = 4$	marginal (renormalizable)	roughly constant
< 0	$d > 4$	irrelevant (non-renormalizable)	grows

Only the last row is dangerous, and for a simple reason: a dimensionless amplitude built from a coupling of dimension $-p$ must carry E^{+p} to compensate. Growth with energy is therefore automatic, not accidental.

For gravity, $[\kappa] = [M]^{-1}$ and the operator $h_{\mu\nu}T^{\mu\nu}$ has dimension 5: gravity sits squarely in the bottom row. The rest of the lecture argues that this is not a catastrophe but a *diagnosis* — and the toy model shows what such a diagnosis normally indicates.

15 A toy model: the Yukawa theory [p. 18*****]

[p. 18*****]

Non-renormalizable interactions: a toy model

Consider the theory described by the Lagrangian density

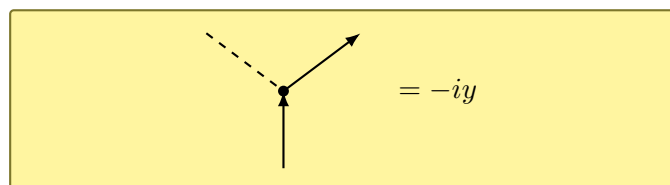
$$\mathcal{L} = \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi + \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \frac{1}{2}M^2\phi^2 - y\phi\bar{\psi}\psi$$

↓ ...interacting with a renormalizable interaction

↓ free Dirac fermion with mass m ; free real scalar field with mass M ...

(a.k.a. “Yukawa interaction”)

↔ The interaction has the following Feynman rule



Let us consider in this theory the process

$$f(p_1) + \bar{f}(p_2) \longrightarrow f(p_3) + \bar{f}(p_4)$$

f and $\bar{f}$ are the fermion and anti-fermion associated to $\psi, \bar{\psi}$

► We identify the following diagrams

Why this particular toy model

The model is engineered to have both a “full” and an “effective” description, so that the relation between them can be seen explicitly:

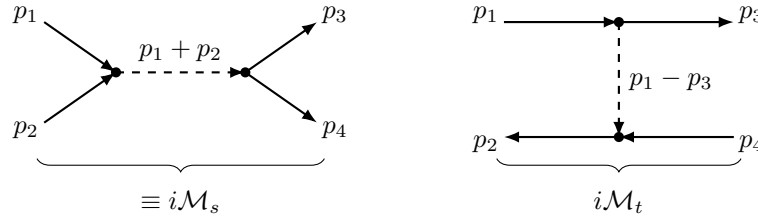
- a *light* fermion ψ of mass m — the degree of freedom we can produce in experiments;
- a *heavy* scalar ϕ of mass $M \gg m$ — too heavy to produce;

- a Yukawa coupling $y\phi\bar{\psi}\psi$, which is perfectly *renormalizable* ($[y] = [M]^0$, operator dimension 4).

Note that the full theory has no problem at all: it is renormalizable and unitary at every energy. The non-renormalizable interaction is going to appear only when we *throw away* the heavy field. That is the entire lesson.

16 The two tree diagrams and their amplitudes [p. 19*****]

[p. 19*****]



We compute the amplitudes

$$\begin{aligned} i\mathcal{M}_s &= \bar{v}(p_2)(-iy)u(p_1) \frac{i}{s - M^2 + i\epsilon} \bar{u}(p_3)(-iy)v(p_4) = \\ &= (-i)y^2 \left(\bar{v}(p_2)u(p_1) \right) \frac{1}{s - M^2 + i\epsilon} \left(\bar{u}(p_3)v(p_4) \right) \end{aligned}$$

$$i\mathcal{M}_s = (-iy^2) \bar{v}(p_2)u(p_1) \frac{1}{s - M^2 + i\epsilon} \bar{u}(p_3)v(p_4)$$

$$\begin{aligned} i\mathcal{M}_t &= \bar{u}(p_3)(-iy)u(p_1) \frac{i}{t - M^2 + i\epsilon} \bar{v}(p_2)(-iy)v(p_4) = \\ &= (-i)y^2 \bar{u}(p_3)u(p_1) \frac{1}{t - M^2 + i\epsilon} \bar{v}(p_2)v(p_4) \end{aligned}$$

$$i\mathcal{M}_t = (-iy^2) \bar{u}(p_3)u(p_1) \frac{1}{t - M^2 + i\epsilon} \bar{v}(p_2)v(p_4)$$

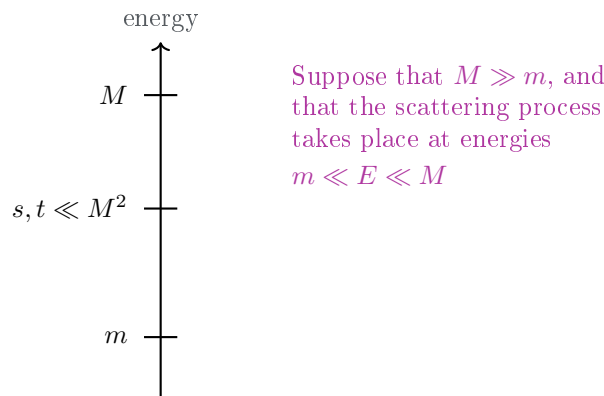
Assembling the amplitudes

The Feynman rules used: $-iy$ at each vertex, the scalar propagator $i/(k^2 - M^2 + i\epsilon)$ on the internal dashed line, and external-leg spinors $u, \bar{u}$ for fermions and $v, \bar{v}$ for antifermions. The Mandelstam variables are $s = (p_1 + p_2)^2$ and $t = (p_1 - p_3)^2$, so the internal momentum squared is s in the first diagram and t in the second.

The factor arithmetic: $(-iy)^2 \times i = (-y^2)(i) = -iy^2$, which is the prefactor in both boxes. Note that the fermion lines are traced through differently in the two diagrams — $\bar{v}(p_2) \dots u(p_1)$ and $\bar{u}(p_3) \dots v(p_4)$ in the s -channel, $\bar{u}(p_3) \dots u(p_1)$ and $\bar{v}(p_2) \dots v(p_4)$ in the t -channel. This pairing is what will need to be matched, current by current, when the two theories are compared on [p. 23*****].

17 The low-energy limit [p. 20*****–p. 21*****]

[p. 20*****]

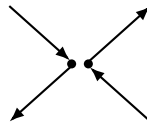


↓ In this situation, we can neglect “ s ” and “ t ” compared to M^2 in the propagator denominator

$$i\mathcal{M}_s^{s,t \ll M^2} = (-iy^2) \bar{v}(p_2)u(p_1) \left(\frac{1}{-M^2} \right) \bar{u}(p_3)v(p_4)$$

$$i\mathcal{M}_s^{s,t \ll M^2} = \left(+\frac{iy^2}{M^2} \right) [\bar{v}(p_2)u(p_1)] [\bar{u}(p_3)v(p_4)] \text{ Eq. (1)}$$

Intuitively, the propagator “shrinks to a point” and we are just left with the fermionic currents



Similarly

[p. 21*****]

$$i\mathcal{M}_t^{s,t \ll M^2} = \left(\frac{iy^2}{M^2} \right) [\bar{u}(p_3)u(p_1)] [\bar{v}(p_2)v(p_4)] \text{ Eq. (2)}$$

What “the propagator shrinks to a point” means

In position space the massive propagator is a Yukawa function $\sim e^{-Mr}/r$: it connects two points only within a distance $1/M$. A process probing distances much larger than $1/M$ — equivalently energies much smaller than M — cannot resolve that separation, and the exchange looks like a single contact interaction.

In momentum space the same statement is the expansion

$$\frac{1}{k^2 - M^2} = -\frac{1}{M^2} \left(1 + \frac{k^2}{M^2} + \frac{k^4}{M^4} + \cdots \right),$$

whose leading term is a constant — no momentum dependence, hence no propagation, hence a point vertex. The neglected terms are suppressed by k^2/M^2 and are exactly the corrections that distinguish the true theory from the contact approximation; keeping them would generate a tower of higher-dimension operators.

This is the single most important mechanism in effective field theory: heavy particles do

not vanish from low-energy physics, they leave behind contact interactions with coefficients suppressed by their mass.

18 The four-fermion effective theory [p. 21*****–p. 22*****]

We now consider the theory described by the Lagrangian density

$$\mathcal{L}_{EFT} = \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi - c(\bar{\psi}\psi)(\bar{\psi}\psi)$$

★

Dirac fermion with non-renormalizable interaction $[c] = [M]^{-2}$

We consider the same scattering process $f(p_1) + \bar{f}(p_2) \rightarrow f(p_3) + \bar{f}(p_4)$ and compute its amplitude using ★

■ When computing the amplitude, there are two kinds of “contractions” that we can do

$$\begin{aligned} \langle f(p_3)\bar{f}(p_4)|(-ic)(\bar{\psi}\psi)(\bar{\psi}\psi)|f(p_1)\bar{f}(p_2)\rangle \times 2 & \quad \begin{aligned} \psi &\sim a u + v b^\dagger \\ \bar{\psi} &\sim a^\dagger \bar{u} + \bar{v} b \end{aligned} \\ = -2ic \left(\bar{u}(p_3)u(p_1)\right) \left(\bar{v}(p_2)v(p_4)\right) & \quad \text{and} \quad = -2ic \left(\bar{v}(p_2)u(p_1)\right) \left(\bar{u}(p_3)v(p_4)\right) \end{aligned}$$

[p. 22*****]

We find

$$i\mathcal{M} = (-2ic) \left[\bar{v}(p_2)u(p_1) \bar{u}(p_3)v(p_4) + \bar{u}(p_3)u(p_1) \bar{v}(p_2)v(p_4) \right] \text{Eq. (3)}$$

Where the two contractions and the factor 2 come from

The operator $(\bar{\psi}\psi)(\bar{\psi}\psi)$ contains four field operators, and each must be contracted with one of the four external particles. The field ψ can annihilate an incoming fermion (the au piece) or create an outgoing antifermion (the $vb^\dagger$ piece); $\bar{\psi}$ does the reverse. The two independent ways of pairing them up give the two terms:

pairing	result
$\bar{\psi}\psi$ makes $\bar{u}(p_3)\dots u(p_1)$, the other makes $\bar{v}(p_2)\dots v(p_4)$	matches the t -channel, Eq. (2)
$\bar{\psi}\psi$ makes $\bar{v}(p_2)\dots u(p_1)$, the other makes $\bar{u}(p_3)\dots v(p_4)$	matches the s -channel, Eq. (1)

The overall $\times 2$ is the symmetry factor from the two *identical* bilinears $(\bar{\psi}\psi)$ in the operator: either factor can supply either pair, and the two choices give the same contribution.

The correspondence with the channels is not a coincidence — it is the statement being set up on the next page.

19 Observation ①: growth with energy [p. 22*****–p. 23*****]

Observation ①

We consider the behaviour of the previous amplitude for energies $E \gg m$

► Since we are only interested in the way the amplitude scales with energy, we will use only dimensional analysis

Since $[u] = [M]^{1/2}$ (same for the other spinors), if we consider energies $E \gg m$ such that we can neglect the mass, the only possibility consistent with dimensional analysis is

$$u, \bar{u}, v, \bar{v} = \mathcal{O}(E^{1/2})$$

The amplitude, therefore, scales according to

[p. 23*****]

$$i\mathcal{M} \sim c E^2$$

Since $[c] = [M]^{-2}$, $i\mathcal{M}$ is dimensionless, as it should be

↔ The amplitude grows with energy, eventually violating unitarity when $i\mathcal{M} \sim \mathcal{O}(1)$

↔ This is expected, since the interaction is non-renormalizable

The same disease as gravity

Compare the two results side by side:

	coupling	amplitude
gravity	$\kappa^2 \sim 1/M_P^2$	$i\mathcal{M} \sim E^2/M_P^2$, breaks at $E \sim M_P$
toy model	$c \sim y^2/M^2$	$i\mathcal{M} \sim y^2 E^2/M^2$, breaks at $E \sim M/y$

Identical structure. But in the toy model we know exactly what is really going on — the four-fermion theory is not fundamental, it is the low-energy shadow of the perfectly healthy Yukawa theory. The apparent breakdown is not a breakdown of physics; it is the point where the approximation stops being valid because a particle we omitted becomes relevant.

That parallel is the whole argument of the lecture, and it is made explicit in Observation 3.

20 Observation ②: matching [p. 23*****]

Observation ②

We note that the amplitude computed with the theory “ $\mathcal{L}_{EFT}$ ” coincides with the low energy limit ($s, t \ll M^2$) of the amplitude computed with the theory “ $\mathcal{L}$ ”

$$\text{Eq. (1)} + \text{Eq. (2)} = \text{Eq. (3)}$$

If we set

$$c \equiv -\frac{y^2}{2M^2}$$

We put forward the following interpretation

The theory

$$\mathcal{L}_{EFT} = \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi + \frac{y^2}{2M^2}(\bar{\psi}\psi)(\bar{\psi}\psi)$$

contains a non-renormalizable interaction and is valid only up to some energy threshold because must be interpreted as the low-energy limit of a renormalizable, “concrete”, theory

Checking the matching, and naming the procedure

Substituting $c = -y^2/2M^2$ into Eq. (3):

$$i\mathcal{M} = (-2ic)[\dots] = \left(+\frac{iy^2}{M^2}\right) \left[\bar{v}(p_2)u(p_1)\bar{u}(p_3)v(p_4) + \bar{u}(p_3)u(p_1)\bar{v}(p_2)v(p_4)\right],$$

whose two terms are exactly Eq. (1) and Eq. (2). The two theories are indistinguishable at energies $E \ll M$.

This procedure — fixing the coefficients of the effective theory by demanding that its amplitudes reproduce those of the full theory at low energy — is called **matching**, and it is how effective theories are constructed in practice.

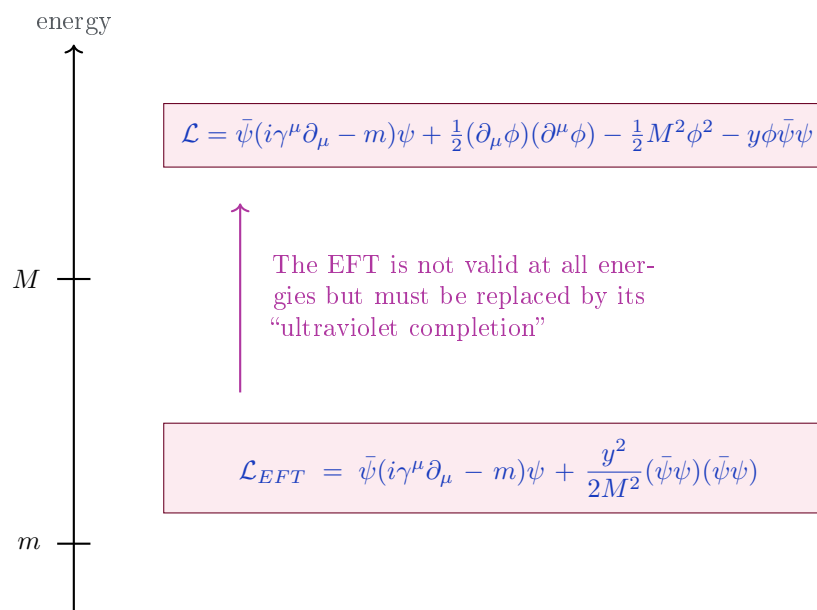
The historical example. This is not an artificial exercise: it is the story of the weak interaction. Fermi wrote a four-fermion contact interaction with a dimensionful coupling $G_F \sim 10^{-5} \text{ GeV}^{-2}$; the amplitude grew like $G_F E^2$ and threatened unitarity around 300 GeV. The resolution was the W boson at 80 GeV, with $G_F/\sqrt{2} = g^2/8M_W^2$ — precisely the relation $c \sim y^2/M^2$ above. Fermi theory is the EFT; the electroweak theory is its ultraviolet completion. The breakdown scale correctly pointed at new physics, and the new physics was found.

21 Effective field theory and ultraviolet completion [p. 24*****—p. 25*****]

[p. 24*****]

A theory of this kind is called an “effective field theory” (EFT)

Schematically:



► At energies $E \ll M$, we can use the EFT. In the EFT, the heavy DOF only appears indirectly via the presence of a non-renormalizable interaction.

The EFT is not valid up to arbitrary high energy since it violates unitarity above some energy scale

↪ Physically, the motivation is now clear.

The EFT does not contain all physical DOF.

At some point, there is enough energy to excite the scalar DOF, which is not contained within [p. 25*****] the EFT.

↪ The energy threshold at which the EFT “breaks down” (a.k.a. EFT cutoff) in this example

is given by

$$E = \mathcal{O}\left(\frac{M}{y}\right)$$

This is a generic result: the EFT cutoff is proportional to the mass of the heavy states and inversely proportional to their coupling with the light DOF.

Why the cutoff is M/y and not just M

Set the amplitude to one: $i\mathcal{M} \sim y^2 E^2 / M^2 \sim 1$ gives $E \sim M/y$. The two limits behave sensibly:

- *Weak coupling* ($y \ll 1$): the cutoff M/y sits *above* M . The EFT formally survives past the mass of the heavy particle — but of course it becomes *wrong* at $E \sim M$, where the heavy particle can be produced on shell. Unitarity violation is a sufficient signal of breakdown, not a necessary one.
- *Strong coupling* ($y \sim 1$): cutoff $\sim M$, and the two coincide.

Either way, the moral holds: *the breakdown scale of an effective theory is an estimate of where new degrees of freedom live*. This is why the growth of an amplitude is informative rather than merely embarrassing — it is a prediction of new physics, and in the case of Fermi theory that prediction was correct.

22 Observation ③: general relativity is an EFT [p. 25*****—p. 26*****]

Observation ③

Going back to gravity.

The fact that interactions involving gravity (gravity with itself, gravity with matter) are non-renormalizable indicate that general relativity must be considered to be an effective field theory

↪ The cutoff is the Planck mass

Contrary to our toy example, however, we do not know what is the ultraviolet completion of gravity

This is both a blessing and a curse

A blessing, because we know GR cannot be the end of the story. A curse, because the energy [p. 26*****]

scale at which we expect to observe new DOF that belong to the UV-completion of gravity is immeasurably large,

$$M_P = 1.22 \times 10^{19} \text{ GeV}$$

$$\leftrightarrow \text{equivalently, } \ell_P \equiv \left(\frac{\hbar G_N}{c^3} \right)^{1/2} \approx 1.68 \times 10^{-35} \text{ m}$$

A numerical slip, and what these numbers mean

The quoted Planck length should be

$$\ell_P = \sqrt{\frac{\hbar G_N}{c^3}} = 1.616 \times 10^{-35} \text{ m} \approx 1.62 \times 10^{-35} \text{ m},$$

not 1.68. (Consistency check: $\ell_P = \hbar/(M_P c)$ with $M_P = 1.22 \times 10^{19} \text{ GeV}$ gives $1.62 \times 10^{-35} \text{ m}$.) The Planck mass quoted is correct.

How far away is this? The LHC reaches about $1.4 \times 10^4 \text{ GeV}$ in the centre of mass — fifteen orders of magnitude below M_P . To probe the Planck scale with a conventional accelerator of current technology one would need a ring of roughly galactic dimensions. This is the “curse”: the theory tells us where it fails, and tells us we cannot go there.

Why it is nonetheless a blessing. A theory that is silently wrong is far more dangerous than one that announces its own limits. General relativity does the latter: it is an excellent effective description below M_P , and it predicts its own replacement above. The programme of this course — start from Lorentz invariance and quantum mechanics, derive a massless spin-2 field, find that consistency forces general relativity, then discover that quantum consistency forces something beyond it — ends here, at the boundary of what is currently known. String theory, asymptotic safety, and loop quantum gravity are proposals for the ultraviolet completion; none is established.

Summary of Lectures 20–21 — the logic in ten lines

Lecture 20 — matter coupled to gravity

1. $m_I = m_G$ means gravity can be removed by going to a freely falling frame — exactly, if the field is constant.
2. Einstein’s principle: for a general field it can still be removed *at each point*, by a set of coordinates that varies from point to point.
3. Recipe: write the physics in flat coordinates, call those coordinates the freely-falling ones, then re-express in arbitrary coordinates.
4. The Jacobian relating the two is the vierbein $e^a_\mu = \partial \xi^a / \partial x^\mu$, and the metric is $g_{\mu\nu} = e^a_\mu e^b_\nu \eta_{ab}$.
5. Applied to a scalar field, the recipe gives minimal coupling: $\eta^{\mu\nu} \rightarrow g^{\mu\nu}$ and $d^4x \rightarrow \sqrt{-g} d^4x$.
6. Expanding $g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}$ with $g^{\mu\nu} = \eta^{\mu\nu} - \kappa h^{\mu\nu}$ and $\sqrt{-g} = 1 + \frac{\kappa}{2} h$ gives $\mathcal{L}_{int} = -\frac{\kappa}{2} h_{\mu\nu} T^{\mu\nu}(\phi)$ — the coupling assumed in Lectures 17–19, now derived.

Lecture 21 — gravity as an effective field theory

7. A $2 \rightarrow 2$ amplitude is dimensionless ($\Delta_{\mathcal{M}} = 4 - n - m$), so $i\mathcal{M} \sim \kappa^2 E^2 = E^2/M_P^2$: it grows with energy and violates unitarity at $E \sim M_P$.

8. This is the generic behaviour of a non-renormalizable interaction — a coupling of negative mass dimension must be compensated by powers of E .
9. The toy model shows what such growth means: integrating out a heavy scalar of mass M from a healthy Yukawa theory leaves a four-fermion operator with $c = -y^2/2M^2$, which reproduces the full amplitude at $E \ll M$ and fails at $E \sim M/y$.
10. Therefore general relativity is an effective field theory with cutoff $M_P = 1.22 \times 10^{19}$ GeV. Unlike the toy model, its ultraviolet completion is unknown — a blessing (GR cannot be final) and a curse (the scale is unreachable).